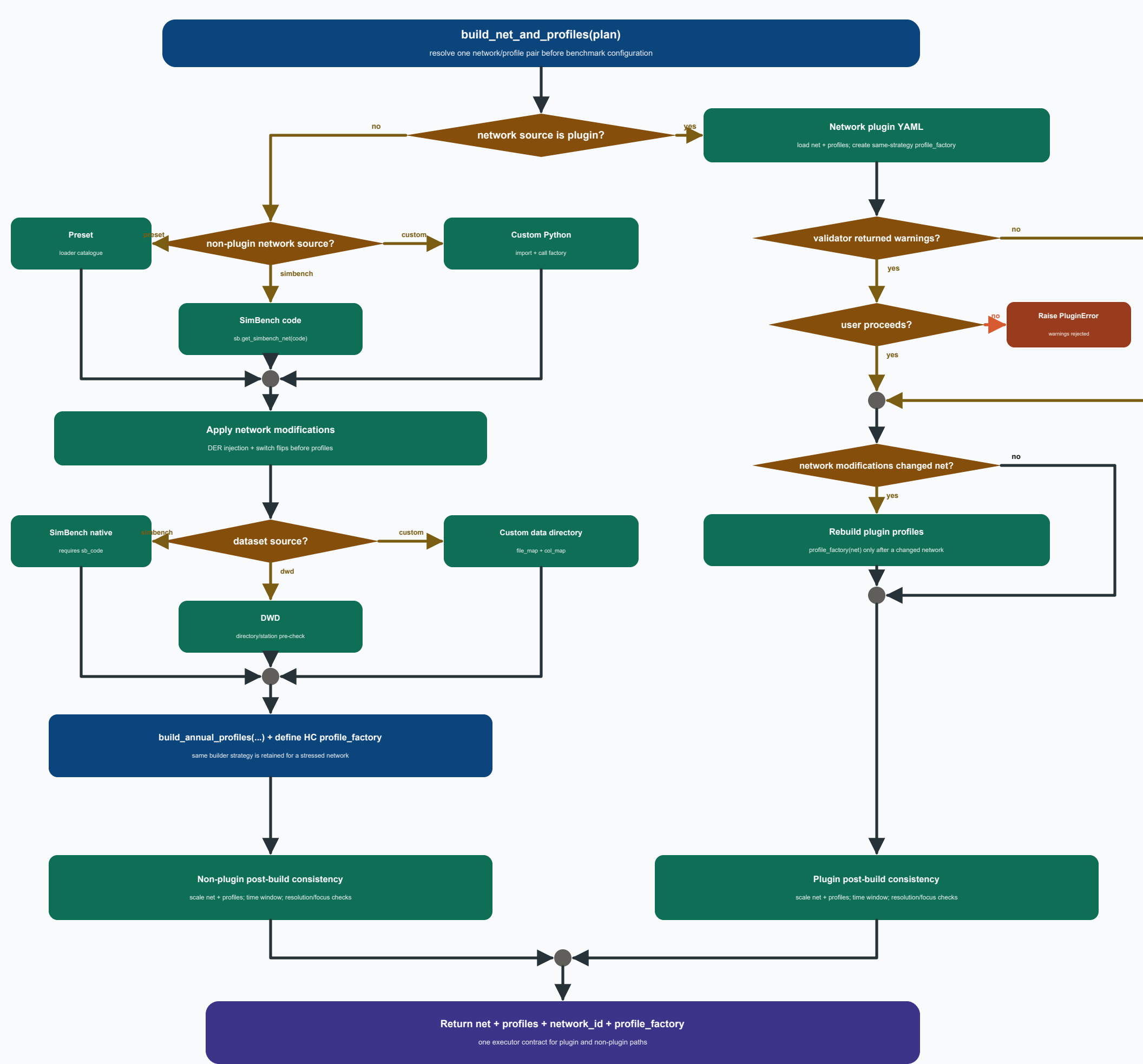




Network resolution

Non-plugin network resolution follows three real source branches: preset loader, assembled SimBench code, or a custom Python factory.

Custom Python networks are imported from the configured file and must return an object with pandapower bus/line/load/sgen tables.

Network-plugin YAML is different: load_network_from_yaml returns net + profiles, then make_profile_factory creates the same-strategy rebuild callable.

Plugin compatibility warnings are shown before execution and require explicit confirmation when any are present.

Dataset resolution

Only non-plugin networks enter dataset dispatch: SimBench native, DWD, or a custom data directory with optional file/column maps.

SimBench-native profiles require an available SimBench code; DWD performs a directory/station pre-check; custom requires custom_path to be a directory.

All three feed profile_builder.build_annual_profiles and create a closure that can rebuild profiles for a later HC-stressed network.

Plugin networks skip this dataset dispatch because their YAML strategy already constructed the initial profile set.

Post-build consistency

User-requested DER injections and switch flips are applied before profile construction or any required plugin-profile rebuild.

Load/DER scaling changes both network tables and matching profile columns; the selected time window is then applied.

timestep_resolution is checked against the actual profile spacing and warns on mismatch; focus_buses are validated against the loaded net.

Both paths return the same four values to the benchmark-configuration phase.